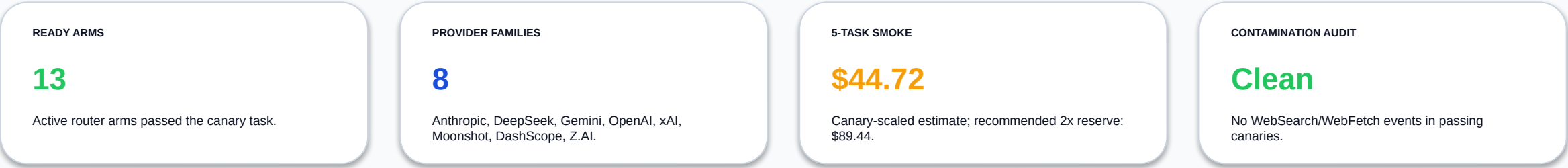
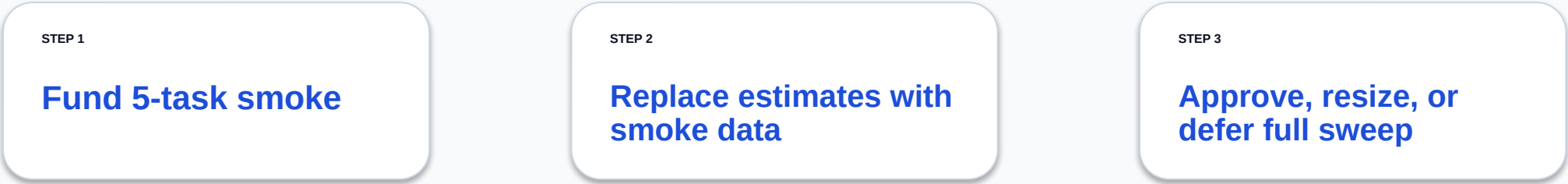


Phase 3 Router Benchmark: Canary-Green, Multi-Provider, Ready for Funded Smoke

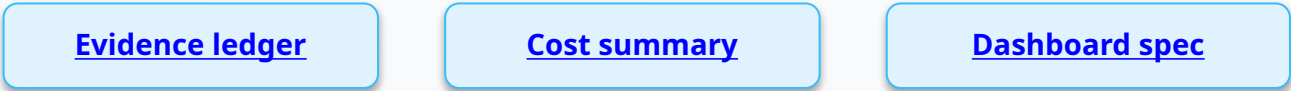
Claude Code remains fixed; LiteLLM routes the same benchmark flow across 13 canary-green arms spanning 8 provider families.



Decision path



Core message: Phase 3 is operational enough for funded smoke testing, but the full sweep should wait until smoke results replace canary-scaled estimates.



From Three-Arm Baseline to Multi-Provider Router Harness

PHASE 1 BASELINE

3 arms

Anthropic Sonnet, DeepSeek Pro, DeepSeek Flash. Claude Code fixed as the agent harness. Harbor ran Terminal-Bench tasks.

PHASE 3 ROUTER LAYER

13 green arms

LiteLLM routes the same benchmark flow across 8 provider families. Canary-first execution catches auth, route, quota, schema, and tool-mode issues.

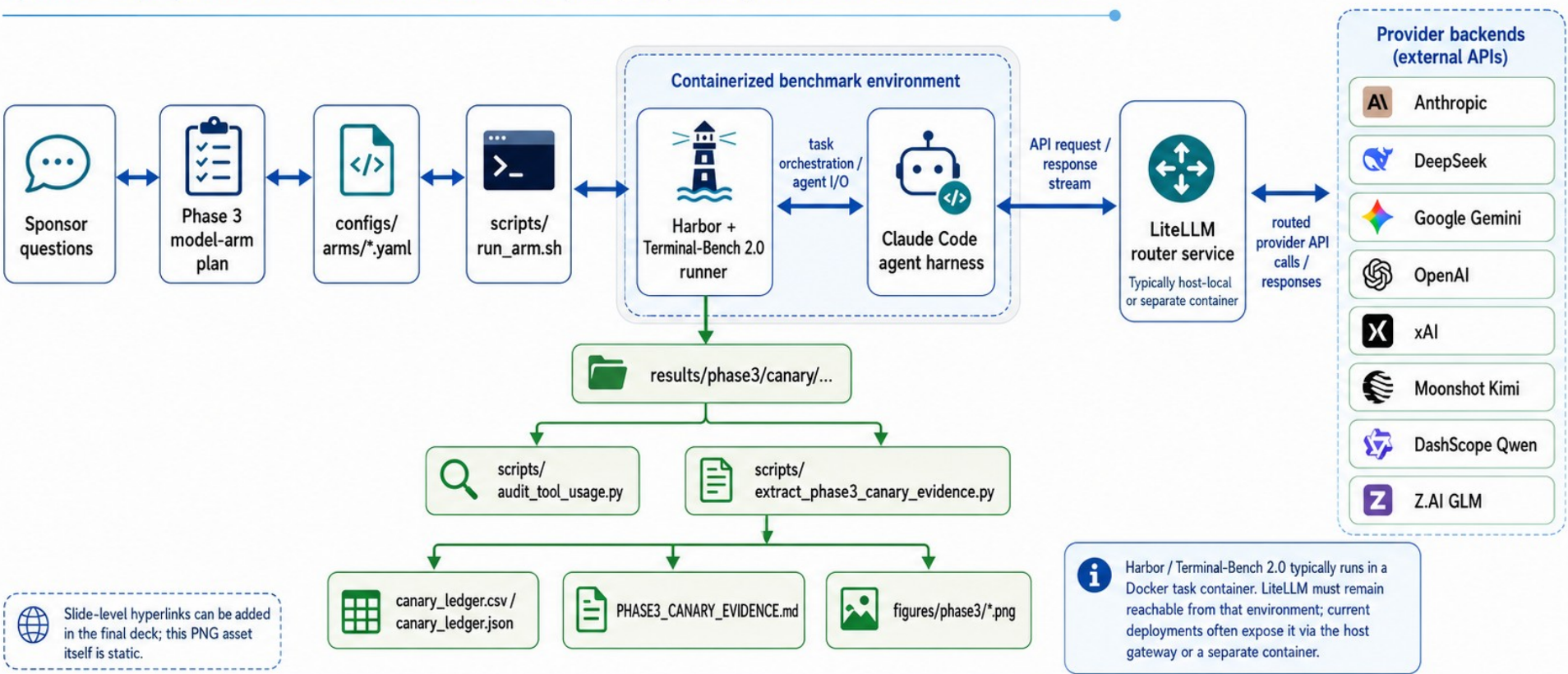
Continuity: Claude Code remains the fixed harness.
Change: model/provider coverage expands through LiteLLM.
Methodology: results remain separated under results/phase3.

Phase 3 Router Benchmark Architecture

Request and response flow is bidirectional between Claude Code, LiteLLM, and provider backends.

Phase 3 Router Benchmark Architecture

Updated deployment view with bidirectional request/response paths and container context



Purpose of canary stage: verify routing, controls, and readiness before funded smoke runs.

Canary Gate: Infrastructure Readiness, Not Final Model Ranking

Current gate uses Terminal-Bench 2.0 task modernize-scientific-stack.

The task exercises read, write, execute, verify, routing, results, and audit flow.

A canary pass means routing/config/tool controls are ready for smoke.

It does not rank model quality across the full Terminal-Bench distribution.

PASS SIGNAL

End-to-end path works

Harbor -> Claude Code -> LiteLLM -> provider -> results -> audit

FAIL SIGNAL

Classify before spending

Auth, quota, route, model slug, schema, tool-mode, or benchmark failure

NEXT GATE

5-task smoke

Use representative multi-task evidence to replace canary-scaled cost estimates

Current Canary-Green Model Arms

13 active router arms spanning 8 provider families.

Arm	Provider	Backend model	Cost	Runtime
router-anthropic-haiku-sanitized	anthropic	claude-haiku-4-5-20251001	\$0.315	486.1s
router-anthropic-opus	anthropic	claude-opus-4-7	\$0.315	289.3s
router-anthropic-sonnet	anthropic	claude-sonnet-4-6	\$0.293	82.3s
router-deepseek-flash	deepseek	deepseek-v4-flash	\$0.115	252.2s
router-deepseek-pro	deepseek	deepseek-v4-pro	\$0.092	425.1s
router-gemini-3.1-pro	google-gemini	gemini-3.1-pro-preview	\$0.291	129.3s
router-gemini-flash	google-gemini	gemini-3.5-flash	\$2.451	150.9s
router-glm-5.1	zai-glm	glm-5.1	\$0.107	135.8s
router-gpt-5.4	openai	gpt-5.4	\$0.799	115.2s
router-gpt-5.5	openai	gpt-5.5	\$2.547	165.1s
router-grok-build-0.1	xai	grok-build-0.1	\$0.710	103.0s
router-kimi-k2.6	moonshot-kimi	kimi-k2.6	\$0.465	198.0s
router-qwen-3.7-plus	dashscope-qwen	qwen3.7-plus	\$0.444	149.8s

Provider-Specific Engineering Lessons

Gemini

Resolved

Route correction and no-plan tool mitigation were needed.

OpenAI

Resolved

Router validation completed; costs need smoke confirmation.

Qwen

Resolved

Singapore endpoint and model-access normalization were required.

xAI Grok

Resolved

Funding/access recovery and model selection unblocked canary.

Moonshot Kimi

Resolved

Temperature/max-token behavior matters for direct probes.

Z.AI GLM

Resolved

Funding/access recovery and model update unblocked canary.

Interpretation: these were infrastructure learnings unless the model actually ran and failed the benchmark task.

Benchmark Contamination Controls

Contamination risk is treated as a first-class control, not an afterthought.



Canary Cost Breakdown

Canary costs reveal budget drivers before smoke/full-sweep funding.



Phase 3 Canary Cost Breakdown

Actual canary costs and scenario-scaled expected costs by active model arm and provider family



Active canary cost sum
\$8.944060



5-task smoke estimate
\$44.72



20 tasks × 3 attempts
\$536.64



25 tasks × 3 attempts
\$670.80



Active canary-green arms

Arm / model	Provider family	Canary \$	5-task smoke \$	20×3 full sweep \$
1. Haiku sanitized	Anthropic	\$0.314650	\$1.57	\$18.88
2. Opus	Anthropic	\$0.315033	\$1.58	\$18.90
3. Sonnet	Anthropic	\$0.292982	\$1.46	\$17.58
4. DeepSeek Flash	DeepSeek	\$0.114800	\$0.57	\$6.89
5. DeepSeek Pro	DeepSeek	\$0.092179	\$0.46	\$5.53
6. Gemini 3.1 Pro	Google Gemini	\$0.291061	\$1.46	\$17.46
7. Gemini Flash	Google Gemini	\$2.451235	\$12.26	\$147.07
8. GLM 5.1	Z.AI GLM	\$0.107285	\$0.54	\$6.44
9. GPT-5.4	OpenAI	\$0.799355	\$4.00	\$47.96
10. GPT-5.5	OpenAI	\$2.546790	\$12.73	\$152.81
11. Grok build 0.1	xAI	\$0.710016	\$3.55	\$42.60
12. Kimi K2.6	Moonshot Kimi	\$0.465169	\$2.33	\$27.91
13. Qwen 3.7 Plus	DashScope Qwen	\$0.443505	\$2.22	\$26.61



Provider family totals

Provider family	Canary \$	5-task smoke \$	20×3 full sweep \$	Share of active canary total
Anthropic	\$0.922665	\$4.61	\$55.36	10.3%
DeepSeek	\$0.206979	\$1.03	\$12.42	2.3%
Google Gemini	\$2.742296	\$13.71	\$164.54	30.7%
Z.AI GLM	\$0.107285	\$0.54	\$6.44	1.2%
OpenAI	\$3.346145	\$16.73	\$200.77	37.4%
xAI	\$0.710016	\$3.55	\$42.60	7.9%
Moonshot Kimi	\$0.465169	\$2.33	\$27.91	5.2%
DashScope Qwen	\$0.443505	\$2.22	\$26.61	5.0%
TOTAL	\$8.944060	\$44.72	\$536.64	100.0%



Largest cost drivers

GPT-5.5 = 28.5% of active canary cost
Gemini Flash = 27.4% of active canary cost
Smoke should test whether these remain persistent budget drivers.



Scenario scaling uses 13 active PASS canaries only. 5-task smoke = ×5; 20×3 full sweep = ×60; 25×3 expanded sweep = ×75.

Add slide-level hyperlinks in the deck; the PNG asset itself is static.

Funding Gate and Forecast

5-TASK SMOKE

\$44.72

Recommended reserve: \$89.44. Run first.

20 TASKS x 3

\$536.64

Reserve: \$804.97. Approve after smoke.

25 TASKS x 3

\$670.80

Reserve: \$1,006.21. Expanded coverage option.

Canary-scaled estimates are useful for order-of-magnitude planning only.
GPT-5.5 and Gemini Flash were high-cost canary outliers.
The ask is to fund the next statistically and financially sensible gate.

Recommended Next Actions

ACTION 1

**Confirm exact 5-task
smoke set**

ACTION 2

**Approve smoke and/or full
sweep budget plus reserve**

ACTION 3

**Run smoke across 13
active green router arms**

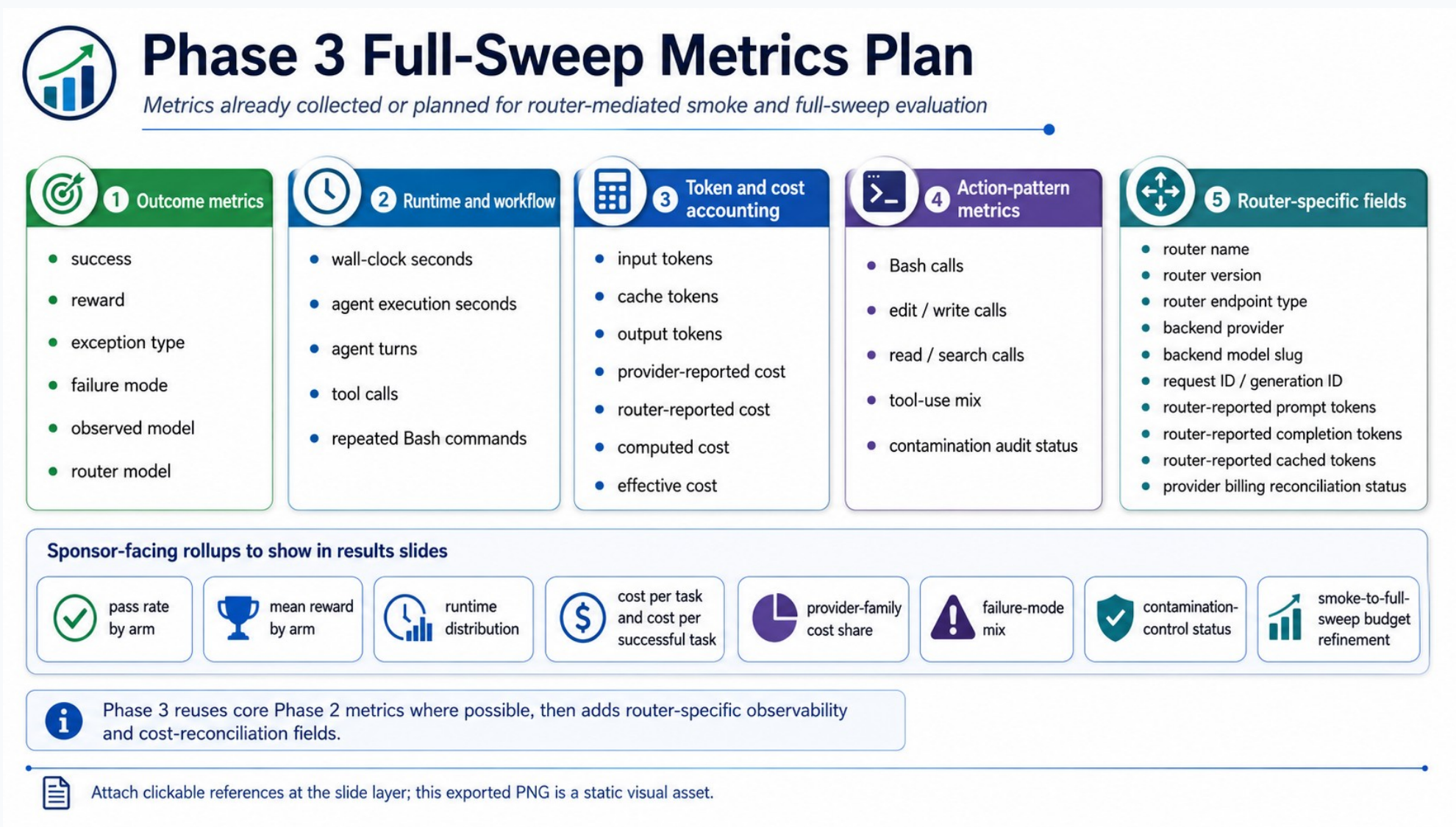
ACTION 4

**Re-run evidence
extraction after smoke**

ACTION 5

**Approve, resize, or defer
full sweep**

Appendix A: Metrics Collected and Planned



Appendix B: Evidence and Reproducibility Hub

[Canary evidence report](#)

[Canary ledger CSV](#)

[Canary ledger JSON](#)

[Cost forecast CSV](#)

[Cost forecast summary](#)

[LiteLLM router config](#)

[Router arm configs](#)

[Contamination notes](#)

[Runbook](#)

[Artifact policy](#)